Lab 2 Reflection

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Technical Description

I did part A of Q2, majority of the coding parts for Q3, and Q5. For Q2.A, I loaded the Excel file and matched the 41 input features from the paper to the dataset's columns. I dropped any rows with missing values and then built X and y that was used for the rest of the lab. For Q3, I built an SVR pipeline with StandardScaler since SVR isn't scale invariant. I used GridSearchCV to tune the kernel and C params. The best model used a linear kernel with C = .1 and I got a testing R2 of .50 and MSE of .29. I repeated this same process with a Ridge pipeline where I tuned alpha instead of kernel and C, which performed very similarly to the SVR. For Q5, I removed 10 random training points to test the stability of SVR vs. Ridge, which showed Ridge to be more stable due to how the model deals with each datapoint. I used Python libraries like pandas, numpy and sklearn for the code.

Positive Critique

Going back to Git from the last lab, during this lab a couple merges went wrong where one accidentally deleted our dataset. Instead of having everyone have to redo their parents, I went back through our commit history and found what was missing and was able to combine everyone's work back together through git versioning. This matters to me because it's showing I'm getting better at using Git and it's experience working through real problems that happen when working with a team.

Negative Critique

Something I want to improve n is planning out how we dish out our questions to start with. Initially before looking at the lab, we all just skimmed through and I said I'd take question 3 without realizing that all the work depended on Q2A for the dataset. And my teammate who was to work on Q2A was working on another part beforehand, so I decided to takeover Q2A, so that we could make the dataset and we could start work on the model building parts. This was a little confusing for the team, and I hope in the future, I will make sure to read through the lab thoroughly once to know what depends on others and be able to split up our work more effectively initially.